CHAPTER 11

Thermodynamics

Thermodynamics deals with the relationship between heat, work, and internal energy. It tells us which processes in nature are possible and which are not. From car engines to refrigerators, from power plants to living cells — all follow the laws of thermodynamics. This chapter covers the zeroth, first, and second laws, thermodynamic processes, and the Carnot engine.

11.1 Introduction

Heat as Energy

SIMPLE EXPLANATION

Thermodynamics (Greek: therme = heat, dynamis = power) is the branch of physics that deals with the conversion of heat into work and vice versa. It studies how energy is transferred between a system and its surroundings.

Before thermodynamics, people thought heat was a fluid called "caloric." Now we know that **heat is a form of energy** — it can be converted to mechanical work (steam engine) and work can be converted to heat (rubbing hands).

KEY TERMS

System: The part of the universe we are studying (e.g., gas in a cylinder).

Surroundings: Everything outside the system.

Boundary: The wall separating system from surroundings (can allow or block heat/matter transfer).

Type of System	Energy Transfer	Matter Transfer	Example
Open	Yes	Yes	Boiling water in open pot
Closed	Yes	No	Gas in a sealed cylinder
Isolated	No	No	Thermos flask (ideal)

11.2 Thermal Equilibrium

DEFINITION

Thermal Equilibrium: Two systems are in thermal equilibrium when they are at the same temperature and there is no net flow of heat between them, even when they are in thermal contact.

SIMPLE EXPLANATION

When you put a hot cup of tea in a room, heat flows from tea to room air. After a long time, the tea reaches room temperature — now tea and room are in thermal equilibrium. No more heat flows.

The key idea: **temperature is the property that decides whether heat will flow or not.** Heat flows only when temperatures are different.

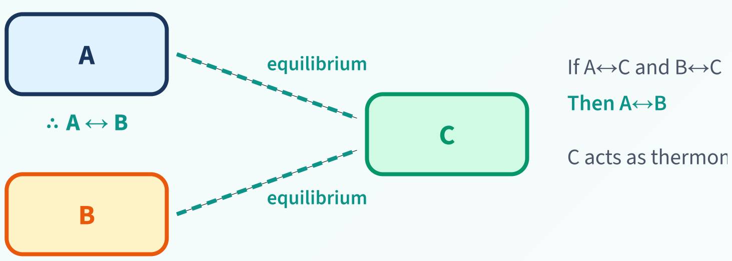
DIAGRAM 11.1: THERMAL EQUILIBRIUM

11.3 Zeroth Law of Thermodynamics

STATEMENT

Zeroth Law: If two systems A and B are each in thermal equilibrium with a third system C, then A and B are also in thermal equilibrium with each other.

DIAGRAM 11.2: ZEROTH LAW



This law defines TEMPERATURE — basis of temperature measurement

WHY IS IT IMPORTANT?

This law is the basis of **temperature measurement**. System C is essentially a **thermometer**. If the thermometer shows the same reading when placed in contact with A and B, then A and B are at the same temperature — even if A and B never touch each other.

It was called "Zeroth" because it was formulated *after* the first and second laws, but is more fundamental — it defines the concept of temperature itself.

11.4 Heat, Work & Internal Energy

(a) Internal Energy (U)

DEFINITION

Internal Energy (U): The total energy contained within a system. It includes the kinetic energy of all molecules (due to their random motion) and the potential energy (due to intermolecular forces).

SIMPLE EXPLANATION

Internal energy is the energy "stored" inside the system. You can't see it directly, but it depends on the **temperature** and **state** of the system.

For an ideal gas: U depends only on temperature ($U = nC_vT$). If temperature increases, U increases.

U is a state function — it depends only on the current state (P, V, T) of the system, not on how the system reached that state.

(b) Heat (Q)

DEFINITION

Heat (Q): The energy transferred between a system and its surroundings due to a temperature difference.

Convention: Q is **positive** when heat is absorbed by the system (heat enters). Q is **negative** when heat is released by the system (heat leaves).

(c) Work (W)

DEFINITION

Work (W): The energy transferred when a system changes its volume against an external pressure.

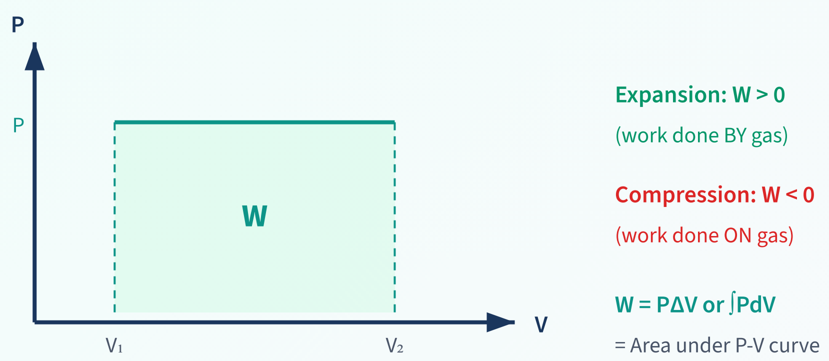
$$W = P\Delta V = P(V_2 - V_1)$$

For expansion ($V_2 > V_1$): W is positive (work done BY system)

For compression ($V_2 < V_1$): W is negative (work done ON system)

General: $W = \int P\,dV$ (area under P-V curve)

DIAGRAM 11.3: WORK DONE BY GAS



HEAT VS WORK — KEY DIFFERENCE

Both Q and W are **path functions** — their values depend on the process (path) taken. But their **difference (Q − W)** is a state function (it equals the change in internal energy).

This is exactly what the First Law of Thermodynamics states!

📌 **Key Points**

- ✓ Internal energy (U): state function, depends only on T for ideal gas
- ✓ Heat (Q): path function; +ve = absorbed, −ve = released
- ✓ Work (W): path function; $W = P\Delta V$; +ve = expansion, −ve = compression
- ✓ W = area under P-V curve

11.5 First Law of Thermodynamics

STATEMENT

First Law: The heat supplied to a system equals the sum of the increase in internal energy and the work done by the system. It is simply the law of conservation of energy applied to thermal systems.

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Q = ΔU + W
Q = heat supplied to system
ΔU = change in internal energy (U_final - U_initial)
W = work done BY the system
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SIMPLE EXPLANATION

Think of it like money: if you receive ₹100 (Q = heat in), you can either save it (ΔU = internal energy increase) or spend it (W = work done). Or you can do both: save ₹60 and spend ₹40.

$Q = \Delta U + W$ means: Heat in = Energy stored + Work done

If no heat enters ($Q = 0$): the system does work at the expense of its internal energy (ΔU is negative, system cools).

Applications of the First Law

Process	Condition	First Law Becomes
Isothermal (const T)	$\Delta U = 0$ (ideal gas)	$Q = W$ (all heat → work)
Adiabatic (no heat)	$Q = 0$	$W = -\Delta U$ (work from internal energy)
Isochoric (const V)	$W = 0$ (no volume change)	$Q = \Delta U$ (all heat → internal energy)
Isobaric (const P)	$W = P\Delta V$	$Q = \Delta U + P\Delta V$
Cyclic process	$\Delta U = 0$ (returns to start)	$Q = W$ (net heat = net work)

EXAMPLE

A gas absorbs 200 J of heat and does 80 J of work. Find the change in internal energy.

$Q = \Delta U + W \longrightarrow 200 = \Delta U + 80$

$\Delta U =$ **120 J** (internal energy increases)

⚠ **COMMON MISTAKES**

- **Mistake:** First law can be violated in special cases
  Correction: First law (energy conservation) is NEVER violated
- **Mistake:** Q and W are state functions
  Correction: Q and W are path functions; only ΔU is a state function

11.6 Specific Heat Capacity of Gases

Gases have two specific heat capacities because they can be heated at constant volume or constant pressure:

DEFINITION

C_v (at constant volume): Heat needed to raise the temperature of 1 mole of gas by 1 K at constant volume. All heat goes to increase internal energy.

C_p (at constant pressure): Heat needed to raise the temperature of 1 mole of gas by 1 K at constant pressure. Heat goes to increase internal energy AND do expansion work.

$$C_p - C_v = R \quad \text{(Mayer's Relation)}$$

R = universal gas constant = 8.314 J/(mol·K)

$C_p > C_v$ always (because extra heat needed for expansion work at constant P)

$$\gamma = C_p/C_v \quad \text{(Heat capacity ratio / Adiabatic index)}$$

Gas Type	C_v	C_p	$\gamma = C_p/C_v$	Example
Monatomic	3R/2	5R/2	5/3 = 1.67	He, Ne, Ar
Diatomic	5R/2	7R/2	7/5 = 1.4	N_2, O_2, H_2
Triatomic+	3R	4R	4/3 = 1.33	CO_2, H_2O

📌 **Key Points**

✓ $C_p > C_v$ always; $C_p - C_v = R$ (Mayer's relation)

✓ $\gamma = C_p/C_v$; γ is always > 1

✓ More atoms in molecule ⟶ more degrees of freedom ⟶ higher C_v ⟶ lower γ

✓ At constant V: $Q = nC_v\Delta T$; At constant P: $Q = nC_p\Delta T$

11.7 Thermodynamic State Variables

DEFINITION

State Variables (State Functions): Properties that depend only on the current state of the system, NOT on how it got there. Examples: Pressure (P), Volume (V), Temperature (T), Internal Energy (U).

STATE VS PATH FUNCTIONS

State functions: P, V, T, U — like your location on a map. Depends only on where you are now, not how you travelled there.

Path functions: Q (heat), W (work) — like distance travelled. Depends on the path taken.

$\Delta U = Q - W$ is a state function even though Q and W individually are path functions!

Equation of State

For an ideal gas, the state variables P, V, and T are related by:

$$PV = nRT$$

This is the equation of state for an ideal gas.
If we know any two of P, V, T — we can find the third.

📌 **Key Points**

- ✓ State variables: P, V, T, U, S (entropy) — depend only on current state
- ✓ Path variables: Q, W — depend on the process taken
- ✓ Equation of state connects P, V, T for a given system
- ✓ A thermodynamic state is completely specified by P, V (for ideal gas, T follows from PV=nRT)

11.8 Thermodynamic Processes

A thermodynamic process is a change of state of a system. Four important processes:

(a) Isothermal Process (Constant Temperature)

DEFINITION

Isothermal Process: A process in which the temperature of the system remains constant throughout. The system is in thermal contact with a heat reservoir.

Condition: $T = \text{constant}$, $\Delta U = 0$ (for ideal gas)
Equation: $PV = \text{constant}$ (Boyle's Law)
Work: $W = nRT \ln(V_2/V_1) = nRT \ln(P_1/P_2)$
First Law: $Q = W$ (all heat converts to work)

(b) Adiabatic Process (No Heat Exchange)

DEFINITION

Adiabatic Process: A process in which no heat enters or leaves the system ($Q = 0$). The system is thermally insulated from surroundings.

$$\text{Condition: } Q = 0$$
$$\text{Equation: } PV^{\gamma} = \text{constant} \quad (\text{also } TV^{\gamma-1} = \text{const})$$
$$\text{Work: } W = (P_1V_1 - P_2V_2)/(\gamma - 1) = nR(T_1 - T_2)/(\gamma - 1)$$
$$\text{First Law: } W = -\Delta U \text{ (work done at the cost of internal energy)}$$

SIMPLE EXPLANATION

Adiabatic expansion: Gas expands and does work → uses its own internal energy → temperature DROPS (gas cools). Example: sudden bursting of a tyre, spray from aerosol can feels cold.

Adiabatic compression: Work is done on gas → internal energy increases → temperature RISES (gas heats up). Example: bicycle pump gets warm, diesel engine ignition.

(c) Isochoric Process (Constant Volume)

DEFINITION

Isochoric Process: A process in which the volume of the system remains constant. No work is done (since $\Delta V = 0$).

$$\text{Condition: } V = \text{constant}, \; W = 0$$
$$\text{First Law: } Q = \Delta U \text{ (all heat} \rightarrow \text{internal energy)}$$
$$Q = nC_v\Delta T$$

Example: Heating gas in a rigid (sealed) container — gas can't expand, so pressure increases.

(d) Isobaric Process (Constant Pressure)

DEFINITION

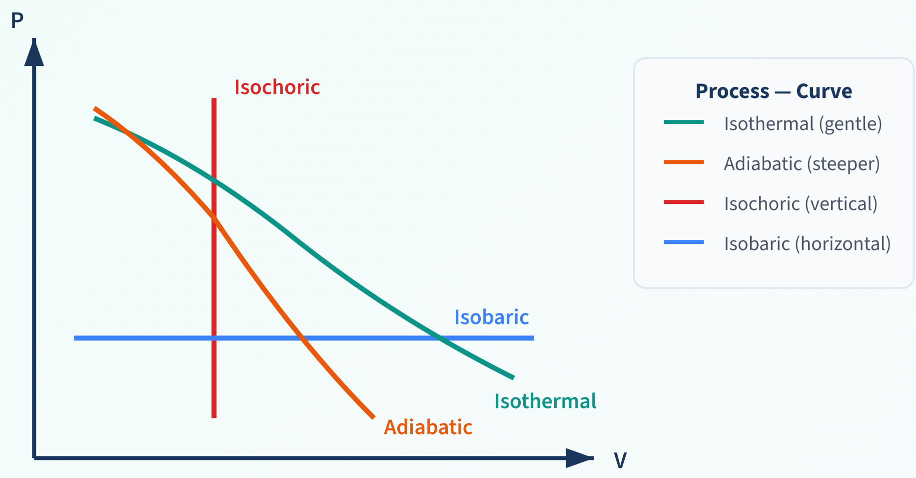
Isobaric Process: A process in which the pressure of the system remains constant.

$$\text{Condition: } P = \text{constant}$$
$$\text{Work: } W = P\Delta V = P(V_2 - V_1) = nR\Delta T$$
$$Q = nC_p\Delta T$$
$$\text{First Law: } Q = \Delta U + P\Delta V$$

Example: Boiling water at atmospheric pressure — water expands into steam at constant P.

P-V Diagrams of All Processes

DIAGRAM 11.4: P-V CURVES FOR ALL FOUR PROCESSES



Process	Constant	Equation	W	Q	ΔU
Isothermal	T	PV = const	$nRT \ln(V_2/V_1)$	W	0
Adiabatic	Q = 0	PV^γ = const	$nR(T_1-T_2)/(\gamma-1)$	0	−W
Isochoric	V	P/T = const	0	$nC_V\Delta T$	Q
Isobaric	P	V/T = const	$P\Delta V = nR\Delta T$	$nC_p\Delta T$	Q − W

EXAMPLE

2 moles of an ideal gas at 300 K expand isothermally to double their volume. Find work done. (R = 8.314)

$W = nRT \ln(V_2/V_1) = 2 \times 8.314 \times 300 \times \ln(2)$

$W = 4988.4 \times 0.693 =$ **3456 J ≈ 3.46 kJ**

📌 **Key Points**

- ✓ Isothermal: T const, ΔU=0, Q=W, slow process in thermal contact
- ✓ Adiabatic: Q=0, W=−ΔU, fast process in insulated system
- ✓ Isochoric: V const, W=0, Q=ΔU, rigid container
- ✓ Isobaric: P const, W=PΔV, $Q=nC_p\Delta T$
- ✓ Adiabatic curve is steeper than isothermal on P-V diagram

11.9 Second Law of Thermodynamics

The first law says energy is conserved. But it doesn't tell us the **direction** of a process. The second law tells us which processes are naturally possible.

KELVIN-PLANCK STATEMENT (ABOUT HEAT ENGINES)

It is impossible to construct a device that operates in a cycle and converts ALL the heat absorbed from a single reservoir completely into work, without any other effect.

Simple language: No engine can be 100% efficient. Some heat must always be rejected.

CLAUSIUS STATEMENT (ABOUT REFRIGERATORS)

It is impossible for heat to flow from a colder body to a hotter body without some external work being done.

Simple language: Heat doesn't flow from cold to hot on its own. A refrigerator needs electricity (work) to transfer heat from inside (cold) to outside (hot).

BOTH STATEMENTS ARE EQUIVALENT

Violating one statement automatically violates the other. Both express the same fundamental truth: **natural processes have a preferred direction**.

- Hot coffee cools down (never spontaneously heats up)
- Gas expands into vacuum (never spontaneously compresses back)
- A ball dropped on floor comes to rest (never spontaneously bounces higher)

📌 **Key Points**

✓ First law: energy is conserved (quantity). Second law: energy has quality (direction)

✓ Kelvin-Planck: no 100% efficient engine possible

✓ Clausius: heat cannot flow cold→hot without external work

✓ All natural processes are irreversible (increase disorder/entropy)

11.10 Reversible & Irreversible Processes

DEFINITION

Reversible Process: A process that can be reversed exactly, passing through the same states in reverse order, leaving no change in the system or surroundings. It is an ideal process — infinitely slow (quasi-static), with no friction or dissipation.

DEFINITION

Irreversible Process: A process that cannot be reversed exactly. The system and surroundings cannot both return to their original states. All real (natural) processes are irreversible.

Property	Reversible	Irreversible
Speed	Infinitely slow (quasi-static)	Finite speed (real process)
Friction	Zero friction	Friction present
Equilibrium	System always in equilibrium	System passes through non-equilibrium states
Efficiency	Maximum possible	Less than maximum
Existence	Ideal (theoretical)	All real processes
Examples	Very slow compression of gas	Burning fuel, mixing, free expansion

WHY STUDY REVERSIBLE PROCESSES?

Reversible processes are ideal and give us the **maximum possible efficiency** for any engine. They set the upper limit. Real engines are always less efficient. The Carnot engine is an example of a theoretically reversible engine.

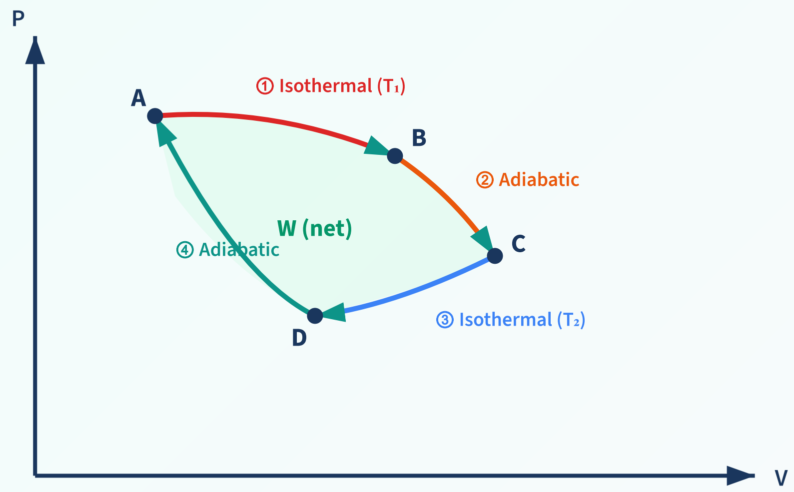
11.11 Carnot Engine

DEFINITION

Carnot Engine: An ideal, reversible heat engine that operates between two temperature reservoirs (hot source and cold sink) using the Carnot cycle. It has the maximum possible efficiency for any engine operating between the same two temperatures.

Carnot Cycle (4 Steps)

DIAGRAM 11.5: CARNOT CYCLE ON P-V DIAGRAM

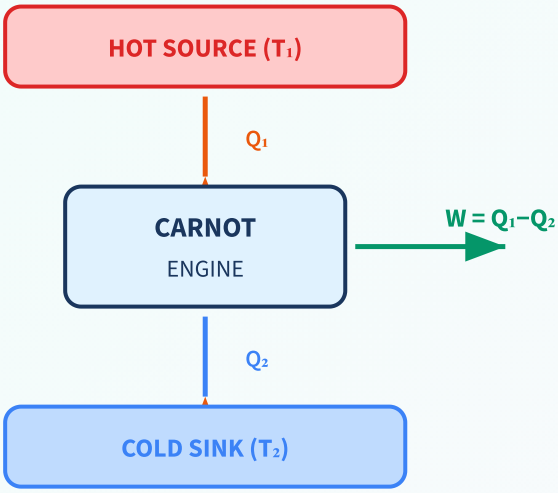


Net work = Area enclosed by cycle | A→B→C→D→A

Step	Process	What Happens
A → B	Isothermal expansion at T_1	Gas absorbs heat Q_1 from hot source, expands slowly
B → C	Adiabatic expansion	Gas expands further, cools from T_1 to T_2 (no heat exchange)
C → D	Isothermal compression at T_2	Gas compressed, releases heat Q_2 to cold sink
D → A	Adiabatic compression	Gas compressed further, heats from T_2 to T_1 (returns to start)

Carnot Engine Schematic

DIAGRAM 11.6: CARNOT ENGINE FLOW DIAGRAM



$\eta = W/Q_1 = 1 - T_2/T_1$

Efficiency of Carnot Engine

$$\eta = 1 - T_2/T_1 = 1 - Q_2/Q_1$$

η = efficiency (always < 1 i.e. < 100%)

T_1 = temperature of hot source (K)

T_2 = temperature of cold sink (K)

T must be in Kelvin!

UNDERSTANDING CARNOT EFFICIENCY

$\eta = 1 - T_2/T_1$ tells us:

- Efficiency increases when T_1 is increased (hotter source)
- Efficiency increases when T_2 is decreased (colder sink)
- η = 100% only if T_2 = 0 K (absolute zero) — which is impossible to achieve
- Therefore, **no engine can ever be 100% efficient** (Second Law!)
- No real engine can be more efficient than a Carnot engine operating between the same temperatures

EXAMPLE 1

A Carnot engine works between 600 K (source) and 300 K (sink). Find efficiency and work done if it absorbs 1000 J.

$\eta = 1 - T_2/T_1 = 1 - 300/600 = 1 - 0.5 = \mathbf{0.5 = 50\%}$

$W = \eta \times Q_1 = 0.5 \times 1000 = \mathbf{500\ J}$

$Q_2 = Q_1 - W = 1000 - 500 = \mathbf{500\ J}$ (rejected to sink)

EXAMPLE 2

A Carnot engine has efficiency 40%. If it rejects 600 J to the sink, find Q_1 and W.

$\eta = 1 - Q_2/Q_1 \rightarrow 0.4 = 1 - 600/Q_1 \rightarrow 600/Q_1 = 0.6$

$Q_1 = 600/0.6 =$ **1000 J**

$W = Q_1 - Q_2 = 1000 - 600 =$ **400 J**

Carnot Refrigerator (Reverse Carnot)

SIMPLE EXPLANATION

A refrigerator is a Carnot engine running in reverse. It uses work (electricity) to transfer heat from cold body (inside fridge) to hot body (room).

$$\text{Coefficient of Performance (COP)} = Q_2/W = Q_2/(Q_1 - Q_2) = T_2/(T_1 - T_2)$$

COP can be > 1 (unlike efficiency which is always < 1)

📌 **Key Points**

- ✓ Carnot engine: ideal reversible engine, maximum possible efficiency
- ✓ $\eta = 1 - T_2/T_1$ (always use Kelvin!)
- ✓ $\eta = 100\%$ only if $T_2 = 0$ K (impossible)
- ✓ No real engine can exceed Carnot efficiency
- ✓ Carnot cycle: isothermal exp → adiabatic exp → isothermal comp → adiabatic comp
- ✓ Refrigerator COP = $T_2/(T_1 - T_2)$

⚡ **Quick Revision Summary - Chapter 11**

- ✓ **Zeroth Law:** Defines temperature; if A↔C and B↔C in equilibrium, then A↔B
- ✓ **Internal Energy (U):** State function; depends on T for ideal gas
- ✓ **Work:** $W = P\Delta V$; W = area under P-V curve; path function
- ✓ **First Law:** $Q = \Delta U + W$ (energy conservation)
- ✓ **$C_p - C_v = R$;** $\gamma = C_p/C_v$ (1.67 mono, 1.4 diatomic)
- ✓ **Isothermal:** T const, PV = const, $\Delta U = 0$, Q = W
- ✓ **Adiabatic:** Q = 0, PV^γ = const, $W = -\Delta U$
- ✓ **Isochoric:** V const, W = 0, $Q = \Delta U = nC_v\Delta T$
- ✓ **Isobaric:** P const, $W = P\Delta V$, $Q = nC_p\Delta T$
- ✓ **Second Law:** No 100% engine (Kelvin); no cold→hot without work (Clausius)
- ✓ **Carnot efficiency:** $\eta = 1 - T_2/T_1$ (maximum for given T_1, T_2)

Practice Questions

Very Short Answer Questions (1 Mark)

1. State the Zeroth Law of Thermodynamics.
2. What is internal energy? Is it a state function or path function?
3. Write the First Law of Thermodynamics.
4. What is an isothermal process?
5. What is an adiabatic process?
6. State the Kelvin-Planck statement of the Second Law.
7. Can a Carnot engine have 100% efficiency? Why not?
8. What is the relation between C_p and C_v?
9. What happens to internal energy in an isothermal process for an ideal gas?
10. What is the work done in an isochoric process?

Short Answer Questions (2-3 Marks)

1. Explain the difference between heat and work as modes of energy transfer.
2. Distinguish between state functions and path functions with examples.
3. State the First Law and explain what happens in each of the four thermodynamic processes.
4. Why is the adiabatic curve steeper than the isothermal curve on a P-V diagram?
5. Explain Mayer's relation $C_p - C_v = R$. Why is $C_p > C_v$?
6. Distinguish between reversible and irreversible processes.
7. State both forms of the Second Law and show they are equivalent.
8. A gas is compressed adiabatically. What happens to its temperature? Explain using the First Law.
9. Why does a bicycle pump get hot when you pump air quickly?
10. What is a Carnot cycle? Name its four steps.

Long Answer Questions (5 Marks)

1. State the First Law of Thermodynamics. Apply it to (a) isothermal, (b) adiabatic, (c) isochoric, and (d) isobaric processes.
2. Describe the Carnot cycle with P-V diagram. Derive the expression for its efficiency.
3. Compare all four thermodynamic processes in a table (constant quantity, equation, W, Q, ΔU) and draw their P-V curves.
4. State and explain both statements of the Second Law. Why can't any engine be 100% efficient?
5. Define C_p and C_v. Derive Mayer's relation. Give the values of γ for monatomic, diatomic, and polyatomic gases.

Numerical Problems

1. A gas absorbs 400 J of heat and does 150 J of work. Find ΔU.
2. In an isothermal process, 3 moles of gas at 300 K expand to twice their volume. Find work done.
3. A Carnot engine operates between 500 K and 300 K. Find its efficiency.
4. A Carnot engine has efficiency 30%. It absorbs 2000 J from the source. Find work done and heat rejected.
5. The temperature of source and sink of a Carnot engine are 227°C and 27°C. Find η. If work output is 500 J, find Q_1.

6. An ideal gas ($\gamma = 1.4$) at 300 K is compressed adiabatically to 1/8 of its volume. Find final temperature.
7. 2 moles of a diatomic gas are heated at constant pressure from 300 K to 500 K. Find Q, W, and ΔU.
8. A refrigerator transfers 500 J from cold reservoir at 250 K to hot reservoir at 300 K. Find work input and COP.
9. In a cyclic process, a gas absorbs 800 J heat and does 500 J work. Find heat rejected in the cycle.
10. Find the efficiency of a Carnot engine whose source is at 127°C and sink at 27°C. If the engine does 1 kJ of work per cycle, find heat absorbed from the source.

Application-Based Questions

1. Why does air escaping from a punctured tyre feel cold? Which thermodynamic process is involved?
2. Explain how a refrigerator works. Does it violate the Second Law?
3. Why can't we power a ship by extracting heat from the ocean water?
4. Why is a diesel engine more efficient than a petrol engine? Relate to Carnot efficiency.
5. Explain why it is easier to cool a room by 5°C using an AC than to heat it by 5°C using reverse cycle.